

Chapter 79

Compositing Layers in Fusion

This chapter is intended to give you a solid base for making the transition from a layer-based compositing application to Fusion's node-based interface. It provides practical information about how to start structuring a node tree for simple layered composites.

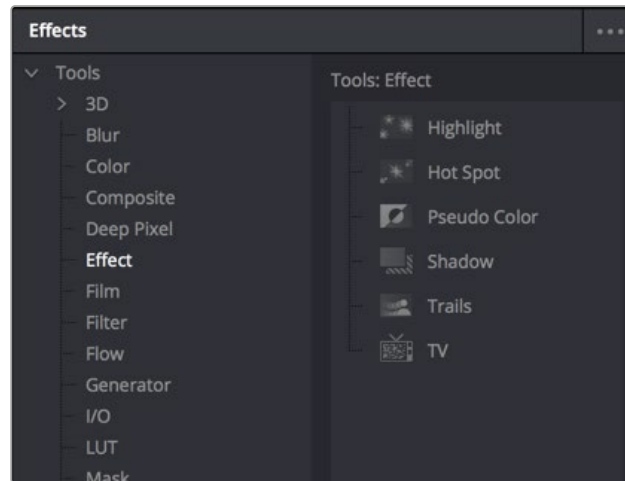
Contents

Applying Effects	1648
Adding a Node to the Tree.....	1648
Editing Parameters in the Inspector.....	1649
Replacing Nodes.....	1650
Adjusting Fusion Sliders.....	1650
Compositing Two Clips Together	1651
Adding Additional Media to Compositions.....	1651
Automatically Creating Merge Nodes.....	1652
Fixing Problem Edges in a Composite.....	1652
Using Composite Modes in the Merge Node.....	1654
Creating and Using Text	1655
Creating Text Using the Text+ Node.....	1655
Styling and Adjusting Text.....	1656
Using Text as a Mask.....	1658
Using Transform Controls in the Merge Node.....	1661
Building a Simple Green-Screen Composite	1662
Mapping Timeline Layers to Nodes in Fusion.....	1662
Pulling a Green-Screen Key Using the Delta Keyer.....	1664
Dealing with Spill.....	1668
Masking a Graphic.....	1668

Applying Effects

Before we dive into multi-layered composites, let's start by looking at some very simple effects and build up from there. Opening the Effects Library, then clicking the disclosure control to the left of Tools, reveals a list of categories containing all the nodes available in Fusion. As mentioned before, each node does one thing, and by using these nodes in concert you can create extremely complex results from humble beginnings.

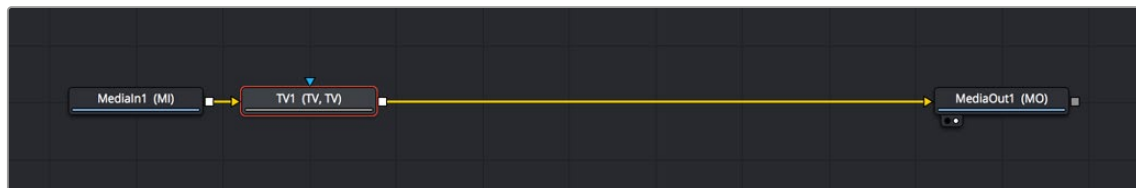
Clicking the Effect category reveals its contents. In this example we'll use the TV effect.



Browsing the Effect category to find the TV node

Adding a Node to the Tree

Assuming that the MediaIn node in the Fusion page or the Loader node in Fusion Studio is the currently selected node in the Node Editor, clicking once on the TV node, for example, in the Effects Library automatically adds that node to the node tree to the right of the selected node. In the Fusion page, it immediately takes effect in the viewer thanks to the fact that the MediaOut1 node is what's loaded in the viewer, since that means that all nodes upstream of the MediaOut1 node will be processed and shown.



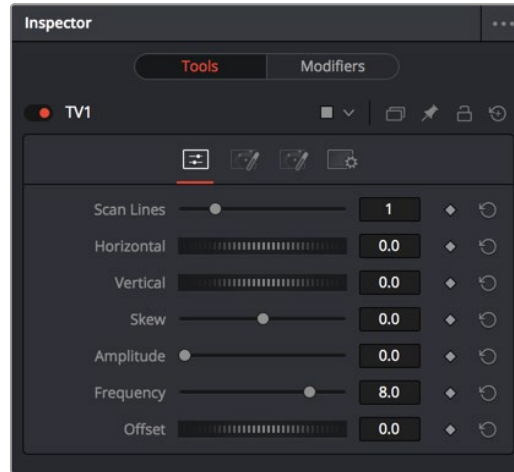
A new node added from the Effects Library

In Fusion Studio, you must press the 1 or 2 key on the keyboard to load the selected node in the viewer.

There are many other ways of adding nodes to your node tree, but it's good to know how to browse the Effects Library as you get started.

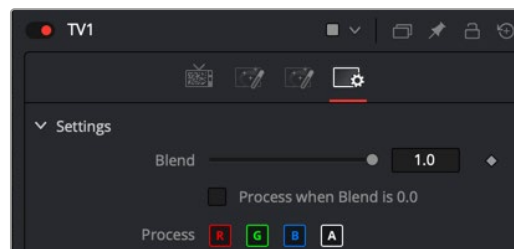
Editing Parameters in the Inspector

To improve an effect, you can make an adjustment to a node's parameters in the Inspector at the right. The selected node shows its controls in the Inspector, in which most nodes have several tabs of controls seen as little icons just underneath that node's title bar.



The Inspector showing the parameters of the TV effect

Clicking the last panel on any node opens the Settings panel. Every node has a Settings panel, and this is where the parameters that every node shares, such as the Blend slider and RGBA buttons, are found. These let you choose which image channels are affected, and let you blend between the effect and the original image.



The Settings panel, which includes channel limiting and mask handling controls that every node shares

In the case of the TV effect, for example, the resulting image has a lot of transparency because the scan lines being added are also being added to the alpha channel, creating alternating lines of transparency. Turning off the Alpha checkbox results in a more solid image, while opening the Controls tab (the first tab) and dragging the Scan Lines slider to the right to raise its value to 4 creates a more visible television effect.



The original TV effect (left), and modifications to the TV effect to make the clip more solid (right)

Replacing Nodes

In the Effect category of the Effects Library, you'll also find a Highlight node that adds glints to the highlights of an image.

Instead of clicking the Highlight node, which would add it after the currently selected node, dragging and dropping a node from the Effects Library on top of a node in the Node Editor replaces the node in the Node Editor.



Dragging a node from the Effects Library onto a node in the Node Editor to replace it

In our example, the Highlight1 node takes the TV node's place in the node tree, and the new effect can be seen in the viewer, which in this example consists of star highlights over the lights in the image.

It's time to use the Inspector controls to customize this effect.

Adjusting Fusion Sliders

When you drag a slider in the Fusion Inspector—in this case, the Number of Points slider—a little dot appears underneath it. This dot indicates the position of the default value for that slider and also serves as a reset button if you click it.



Adjusting a slider reveals a reset button underneath it

Each slider is limited to a different range of minimum and maximum values that is particular to the parameter you're adjusting. In this case, the Number of Points slider maxes out at 24. However, you can remap the range of many (but not all) sliders by entering a larger value in the number field to the right of that slider. Doing so immediately repositions the slider's controls to the left as the slider's range increases to accommodate the value you just entered.



Entering a larger value to expand the range over which a slider will operate

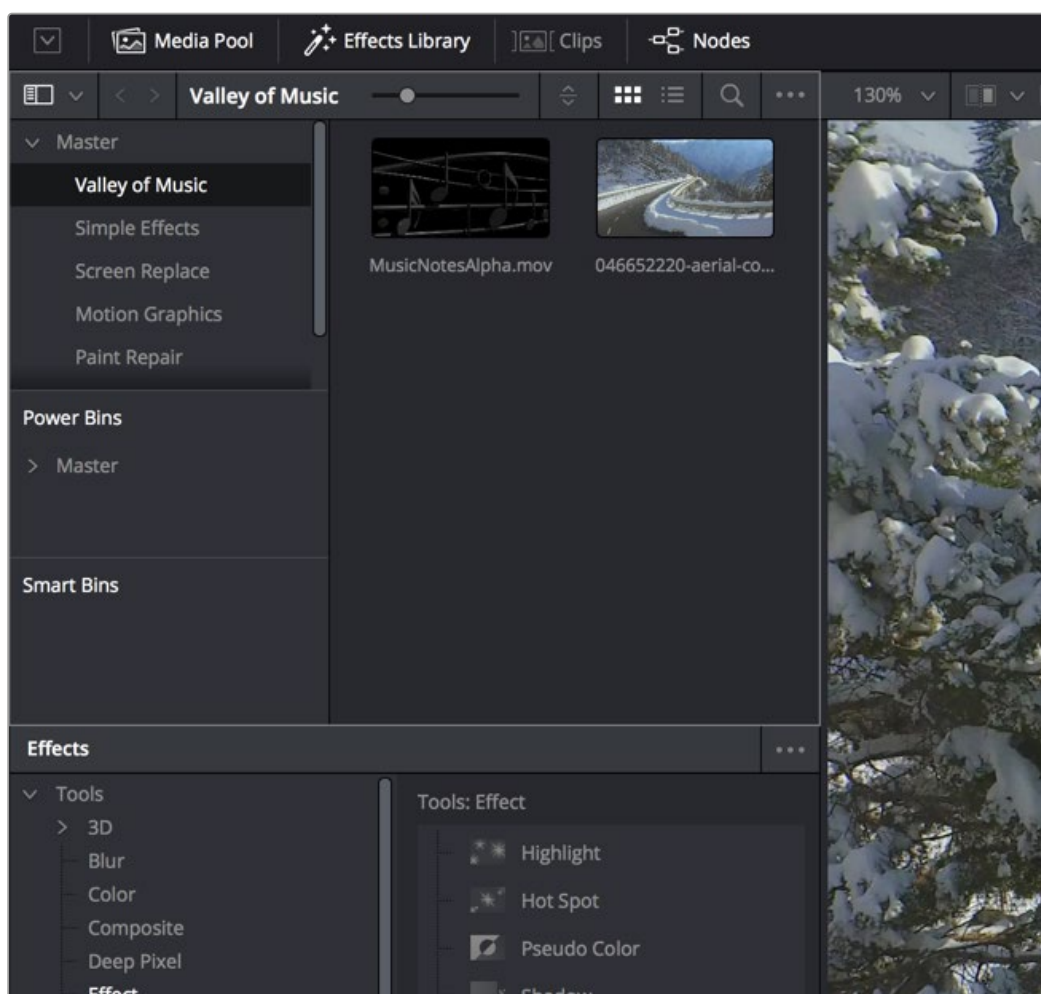
Compositing Two Clips Together

As entertaining as it is adding individual nodes to create simple effects, eventually you'll need to start adding additional layers of media in order to merge them together as composites. Let's turn our attention to another example in which we need to combine a background clip with a foreground clip that includes a built-in alpha channel, to see simple layering in action.

Adding Additional Media to Compositions

You'll often find that even though you start out wanting to do something relatively simple, you end up adding additional media to create the effect that you need.

- In Fusion Studio, you do this by adding additional Loader nodes. If you add a new Loader node to an empty area of the Node Editor, you'll add an unconnected Loader2 node (incremented to keep it unique) that you can then connect how you want.
- In the Fusion page, you can open the Media Pool and drag clips directly to the Node Editor to add them to your node tree. If you drag a clip from the Media Pool to an empty area of the Node Editor, you'll add an unconnected MediaIn2 node (incremented to keep it unique) that you can then connect in any way you want.

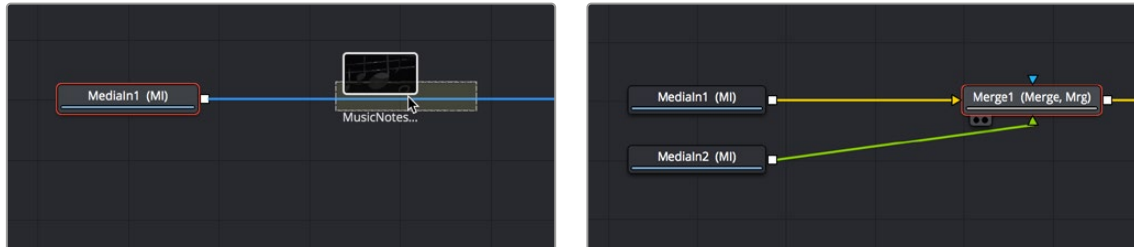


The Media Pool as seen in the Fusion page

Automatically Creating Merge Nodes

If you want to connect the incoming clip immediately to your node tree as the top layer, or foreground, of a composite, in Fusion Studio select the Loader1 node and then add a second Loader node. In the Fusion page, drag the new clip from the Media Pool right on top of any connection line.

In both cases, the new MediaIn or Loader node automatically becomes the “foreground input”.



Dragging a node from the Media Pool onto a connection (left), and dropping it to create a Merge node composite (right)

The Node Editor is filled with shortcuts like this to help you build your compositions more quickly. Here's one for when you have a disconnected node that you want to composite against another node with a Merge node. Drag a connection from the output of the node you want to be the foreground layer, and drop it on top of the output of the node you want to be the background layer. A Merge node will be automatically created to build that composite. Remember: background inputs are orange, and foreground inputs are green.



Dragging a connection from a disconnected node to another node's output (left), and dropping it to create a Merge node composite (right)

Adding Clips to a Fusion Composition From the File System

If you drag clips from the file system directly into the Node Editor, they'll be added to the DaVinci Resolve Media Pool automatically. So, if you have a library of stock animated background textures and you've just found one you want to use using your file system's search tools, you can simply drag it straight into the Node Editor and it'll be added to the currently selected bin of the Media Pool.

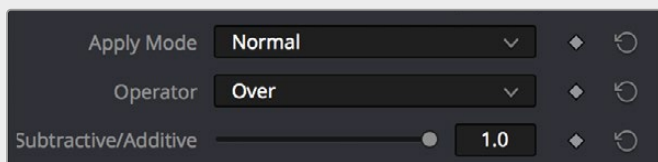
Fixing Problem Edges in a Composite

Most often, the Merge node does a perfectly good job when handed a foreground image with premultiplied alpha transparency to composite against a solid background image. However, from time to time, you may notice a small bit of fringing at the edge of the border of a foreground element and transparent area, such as seen in the following close-up. This slight lightening at the edge is a tell-tale sign that the clip probably wasn't premultiplied. The Merge node expects all foreground images with alpha channels to be premultiplied. But this is something that's easily fixed.



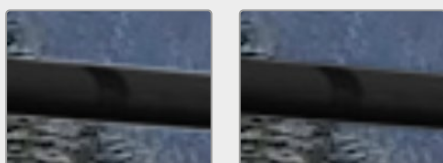
A bit of fringing at the edge of a foreground element surrounded by transparency

Click to select the Merge node for that particular composite, and look for the Subtractive/Additive slider.



The Subtractive/Additive slider, which can be used to fix or improve fringing in composites

Drag the slider all the way to the left, to the Subtractive position, and the fringing disappears.



A clip with alpha exhibits fringing (left), and after fixing fringing by dragging the Subtractive/Additive slider to the left (right)

The Subtractive/Additive slider, which is only available when the Apply mode is set to Normal, controls whether the Normal mode performs an Additive merge, a Subtractive merge, or a blend of both. This slider defaults to Additive merging, which assumes that all input images with alpha transparency are premultiplied (which is usually the case). If you don't understand the difference between Additive and Subtractive merging, here's a quick explanation:

- An Additive merge, with the slider all the way to the right, is necessary when the foreground image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the alpha channel. The result is that transparent pixels are always black, since any number multiplied by 0 is always 0. This obscures the background (by multiplying with the inverse of the foreground alpha), and then simply adds the pixels from the foreground.
- A Subtractive merge, with the slider all the way to the left, is necessary if the foreground image is not premultiplied. The compositing method is similar to an Additive merge, but the foreground image is first multiplied by its own alpha to eliminate any background pixels outside the alpha area.

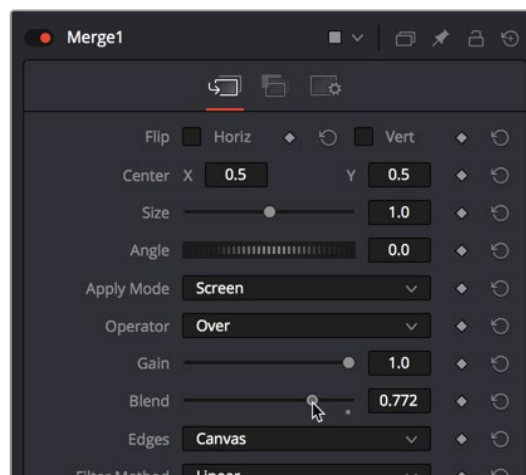
The Additive/Subtractive slider lets you blend between two versions of the merge operation, one Additive and the other Subtractive, to find the best combination for the needs of your particular composite. Blending between the two is occasionally useful for dealing with problem composites that have edges that are calling attention to themselves as either too bright or too dark.

For example, using Subtractive merging on a premultiplied image may result in darker edges, whereas using Additive merging with a non-premultiplied image will cause any non-black area outside the foreground's alpha to be added to the result, thereby lightening the edges. By blending between Additive and Subtractive, you can tweak the edge brightness to be just right for your situation.

Using Composite Modes in the Merge Node

To create a more convincing composite in layer-based systems, you often use Blend modes. Blend modes are located in the Merge node since that is where one layer gets composited over another. Let's take an example where you want to use the Screen mode to make a foreground image look more like a reflection.

The Merge node has a variety of controls built into it for creating just about every compositing effect you need. Items you may be familiar with as Blend modes are located in the Apply Mode pop-up menu. You can use these mathematical compositing modes to combine the foreground and background layers together. A Blend slider allows you to fade the foreground input with the background.



Adjusting the Apply Mode and Blend slider of the Merge node in the Inspector

NOTE: The Subtractive/Additive slider disappears when you choose any other Apply Mode option besides Normal, because the math would be invalid. This isn't unusual; there are a variety of controls in the Inspector that hide themselves when not needed or when a particular input isn't connected.

The Screen node is perfect for simulating reflections, and lowering Blend a bit lets you balance the foreground and background images. It's subtle, but helps sell the shot.

TIP: You may have noticed that the Merge node also has a set of Flip, Center, Size, and Angle controls that you can use to transform the foreground image without needing to add a dedicated Transform node. It's a nice shortcut for simplifying node trees large and small.

Creating and Using Text

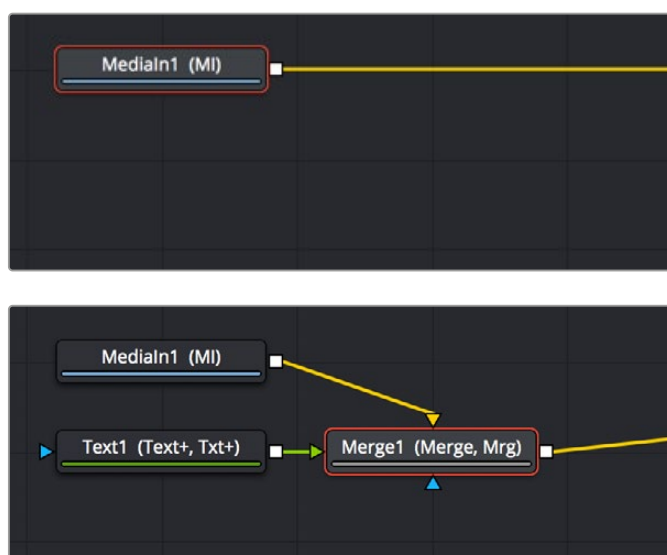
In this next example, we'll look at how to create a simple text object using the Text+ node. Then, we'll see how to use the text generator's alpha channel in another image to create a more complex composite.

Creating Text Using the Text+ Node

The Text+ node is the primary tool for creating 2D text in the Fusion page. If you are using DaVinci Resolve, this is also the same Text+ generator available in the Edit page. It is easily accessible right in the toolbar.

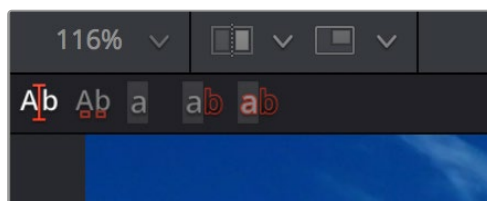
The Text+ node is an incredibly deep tool for creating text effects, with six tabs of controls for adjusting everything from text styling, to different methods of layout, to a variety of shading controls including fills, outlines, shadows, and borders. As sophisticated a tool as this is, we'll only be scratching the surface in this example.

We'll start with a MediaIn node that will serve as our background selected in the Node Editor. Clicking the Text+ button automatically creates a new Text+ node connected as the foreground input of a Merge node. The same behavior occurs if you are using Fusion Studio, with a Loader node.



Selecting a node you want to append another node to (top) Clicking the Text+ button on the toolbar automatically creates a Merge composite with the text as the foreground input connection (bottom)

Selecting a Text node opens the default Text panel parameters in the Inspector, and it also adds a toolbar at the top of the viewer with tools specific to that node. Clicking on the first tool at the left lets you type directly into the viewer, or you can type into the Styled Text field in the Inspector.



The viewer toolbar for the Text node with tools for text entry, kerning, and outline controls

If you're viewing the Merge, the text appears in the viewer superimposed against the background clip. Onscreen controls appear that let you rotate (the circle) and reposition (the red center handle and two arrows) the text, and we can see a faint cursor that lets us edit and kern the text using other tools in the viewer toolbar.



Text that's been typed into the viewer, with onscreen text transform controls

Styling and Adjusting Text

To style the text, you use the controls in the Inspector, modifying text style controls such as Font, Size, and Tracking to change the spacing between the letters.

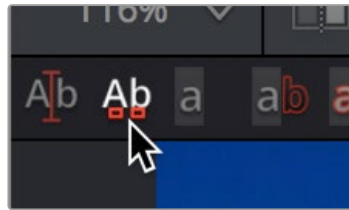


The restyled text

TIP: Holding down the Command key while dragging any control in the Inspector “gears down” the adjustment so that you can make smaller and more gradual adjustments.

Selecting the Manual Kerning tool in the viewer toolbar (second tool from the left) reveals small red dots underneath each letter of text.

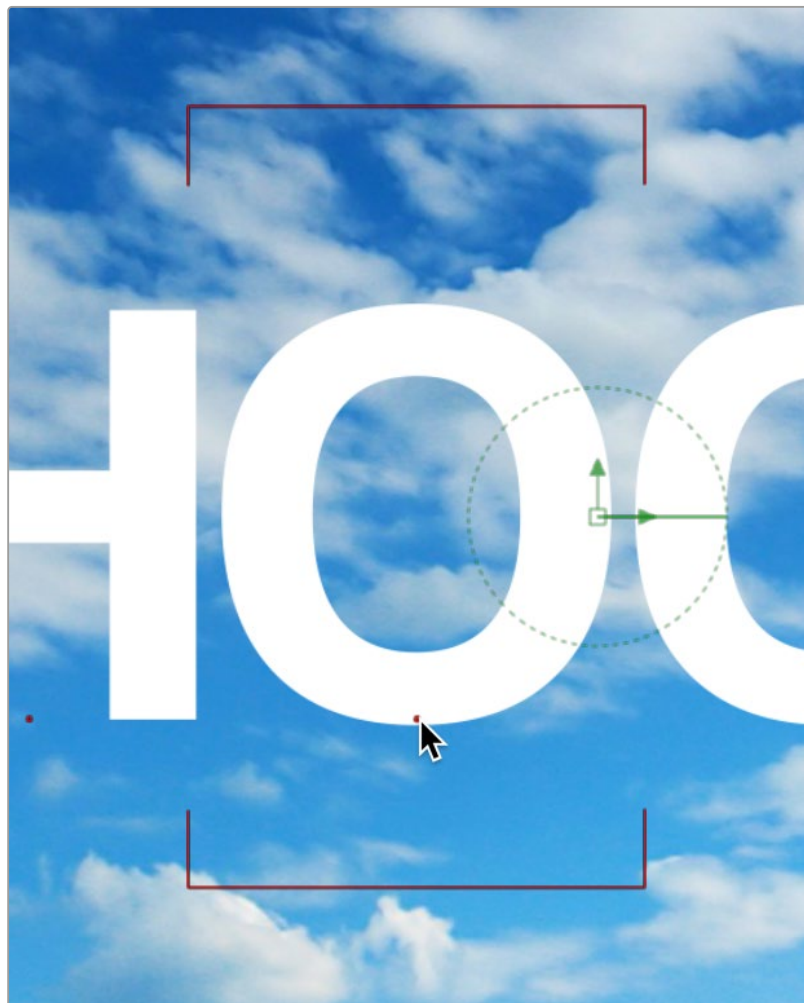
Clicking a red dot under a particular letter puts a kerning highlight over that letter.



The Manual Kerning tool
in the viewer toolbar

To make manual kerning adjustments:

- 1 Option-drag the red dot under any letter of text to adjust that character's kerning while constraining letter movement to the left and right. You can also drag letters up and down for other effects. Depending on your system, the kerning of the letter you're adjusting might not update until you drop the red dot in place.
- 2 If you don't like what you've done, you can open the Advanced Controls in the Inspector and clear either the kerning of selected letters or all manual kerning before starting over again.



Option-dragging the little red dot revealed by the Manual
Kerning tool to manually adjust kerning

Using Text as a Mask

You can fill the text with a color or gradient, or you can use the text as a matte to cut the letters out of another image. First, we'll drag another clip of a chalkboard covered with math from the Media Pool to the Node Editor as a disconnected MediaIn2 node.

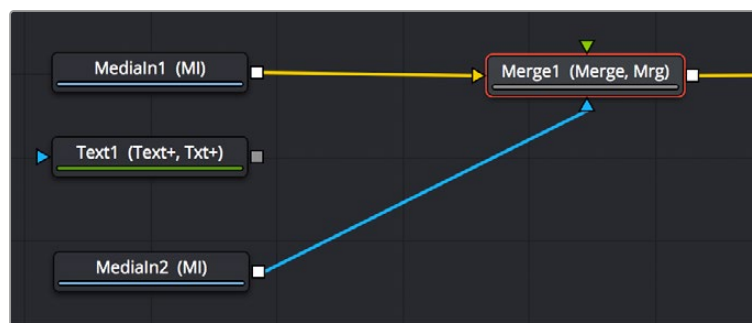
Disconnecting and Reconnecting Nodes

If we start from the previous example that had a MediaIn as the background and the Text+ as the foreground to a Merge, we'll need to do a little rearranging. Clicking the last half of the connection from the Text1 node to the Merge foreground input disconnects it.



Clicking the second half of a connection to disconnect it (previous page), and the result with the text node disconnected (above)

Connecting a MediaIn2 or Loader2 node onto the Merge1 node's foreground input causes the entire viewer to be filled with the MediaIn2 (assuming we're still viewing the Merge node). At this point, we need to insert the Text1 node's image as an alpha channel into the MediaIn2 node's connection, and we can do that using a MatteControl node.

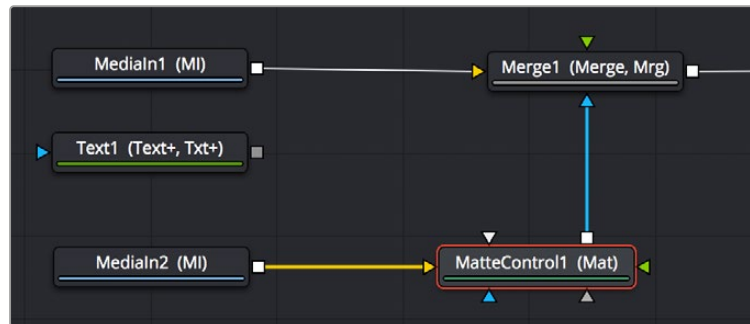


The updated composite, with two video images connected and the text node disconnected

Using Matte Control Nodes

Select the MediaIn2 node, and click the Matte Control button in the toolbar to add it between the MediaIn2 and Merge1 nodes. (To tidy up, I've moved the nodes around a bit in the screenshot.)

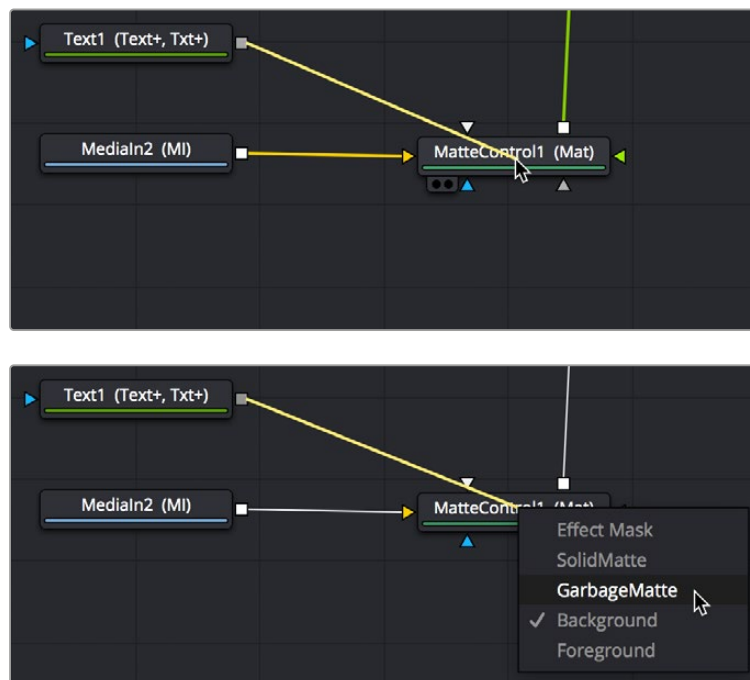
The MatteControl node has numerous uses. Among them is taking one or more alpha channels, mattes, or images that are connected to the Garbage Matte, Solid Matte, and/or foreground inputs, combining them, and using the result as an alpha channel for the image that's connected to the background input. It's critical to make sure that the image you want to add an alpha channel to is connected to the background input of the MatteControl node, or the MatteControl node won't work.



The second image properly connected to the MatteControl node's background input

With this done, connecting the Text+ node's output, which has the alpha channel, to the MatteControl node's Garbage Matte input, is a shortcut we can use to make a mask, matte, or alpha punch out a region of transparency in an image.

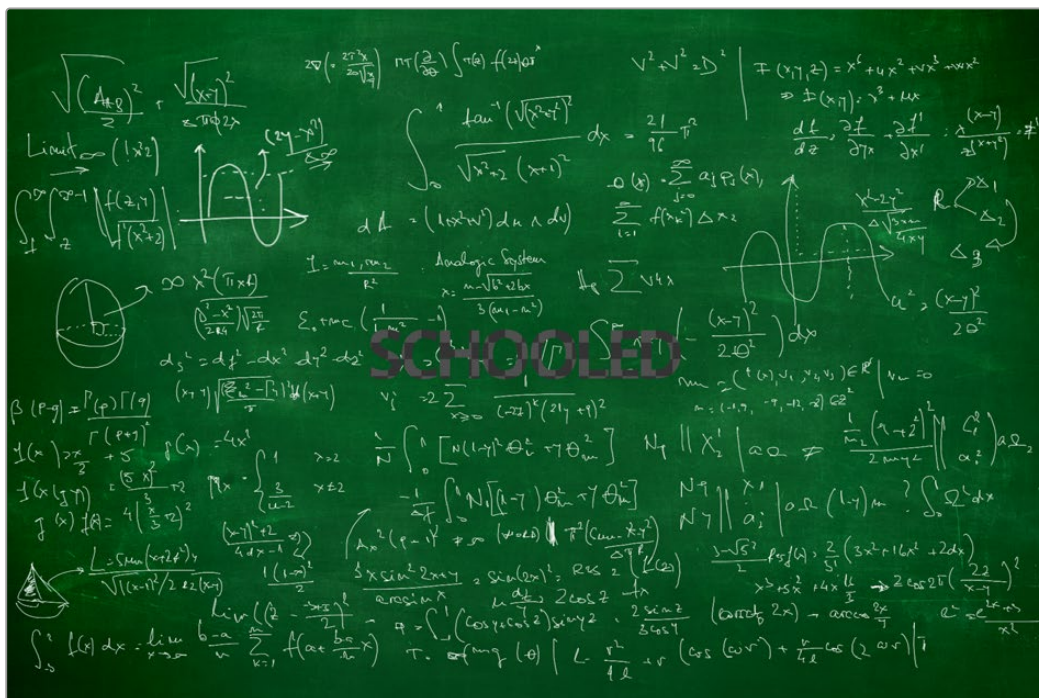
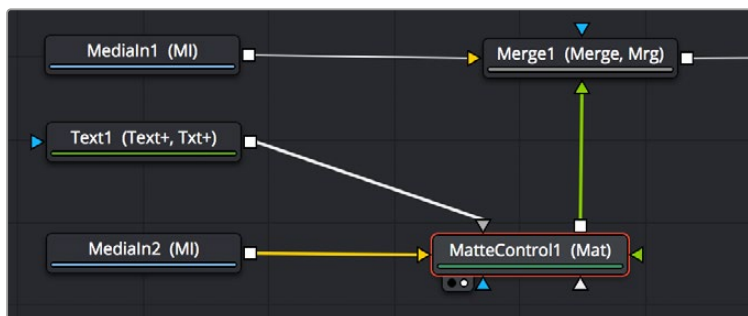
Keep in mind that it's easy to accidentally connect to the wrong input. Because inputs rearrange themselves depending on what's connected and where the node is positioned (and, frankly, the colors can be hard to keep track of when you're first learning), it's key to make sure that you always check the tooltips associated with the input you're dragging a connection over to make sure that you're really connecting to the correct one. If you don't, the effect won't work, and if your effect isn't working, the first thing you should always check is whether you've connected the proper inputs.



Option-dragging a node connection to drop onto another node exposes a node input menu

One alternate method of connecting nodes together is to hold down the Option key while dragging a connection from one node's output and dropping it onto the body of another node. This opens a pop-up menu from which you can choose the specific input you want to connect to, by name. Note that the menu only appears after you've dropped the connection on the node and released your pointing device's button.

Once the Text1 node is properly connected to the MatteControl node's Garbage Matte input, a text-shaped area of transparency is displayed for the graphic if you load the MatteControl node into the viewer.



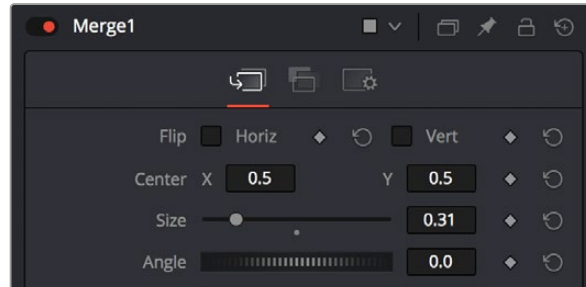
Connecting the Text node to the MatteControl node's Garbage Matte input (previous page), and the resulting hole punched in the image (above)

Customizing Matte Control Nodes

You can use the Inspector to change some parameters to get the result you want. In the Inspector controls for the MatteControl node, revealing the Garbage Matte controls exposes parameters for modifying how the Garbage Matte input is applied to the image. For example, you can choose to have the text mask filled with the image instead of cutting a hole in the image.

Using Transform Controls in the Merge Node

The Merge node includes a set of transform parameters in the Inspector that specifically affect the foreground input's image. This makes it quick and easy to adjust a foreground image to match the background without requiring another node.



The Merge node transform controls that affect the foreground input's image

NOTE: When connecting two images of different sizes to a Merge node, the resolution of the background image defines the output resolution of that node. Keep that in mind when you run into resolution issues.

You can use the Size slider to resize the foreground.



The final composite

Building a Simple Green-Screen Composite

Taking another step forward in compositing, the next example shows how you can equate a multilayered Timeline like the one in DaVinci Resolve's Edit page to nodes in Fusion's node tree. We'll use DaVinci Resolve, but understanding how layers map to nodes can be helpful for anyone new to dealing with a node-based interface. In this example, we'll create a simple composite using a green-screen key and two other layers to create a news story.

Mapping Timeline Layers to Nodes in Fusion

This composite involves three layers in a Timeline. The Timeline consists of a background graphic on video track 1, a green-screen clip on video track 2, and a foreground graphic on video track 3.

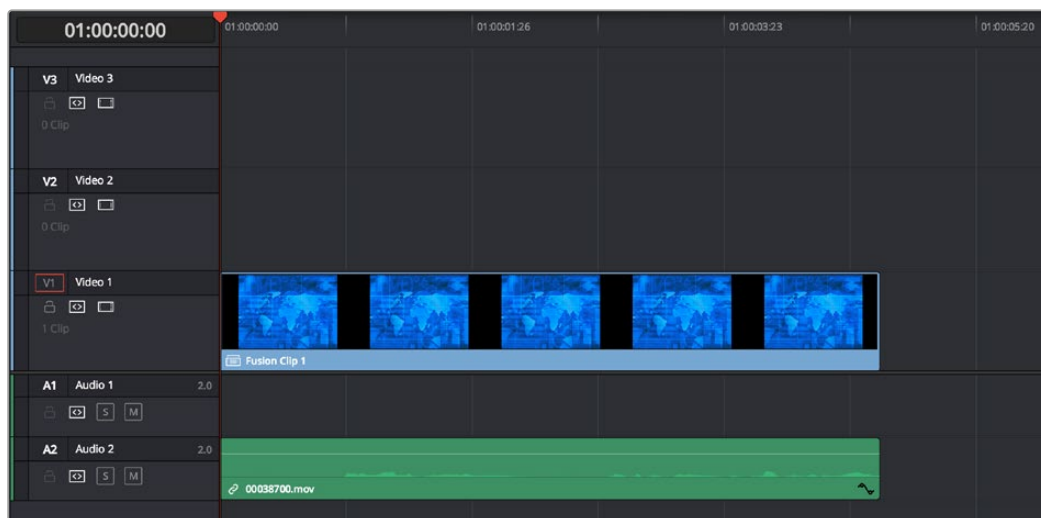
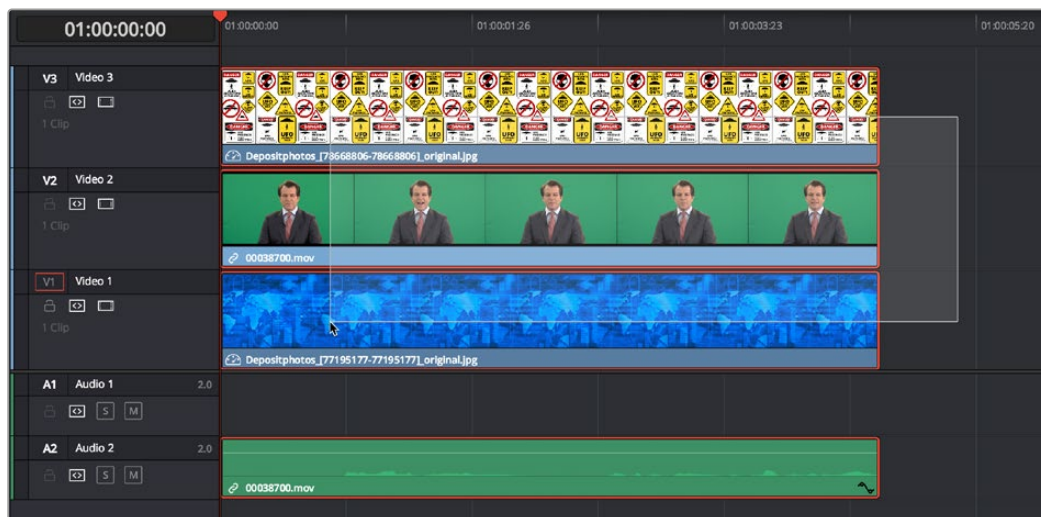


Background on video track 1 (top left), green-screen clip on video track 2 (bottom), and graphic file on video track 3 (top right)

Implied in a timeline-based system is that higher numbered video tracks appear as the more forward, or frontmost, element in the viewer. Video track 1 is the background to all other video tracks. Video track 3 is in the foreground to both video track 1 and video track 2.

TIP: If using DaVinci Resolve, you can bring all three layers from the Edit page into Fusion by creating a Fusion clip.

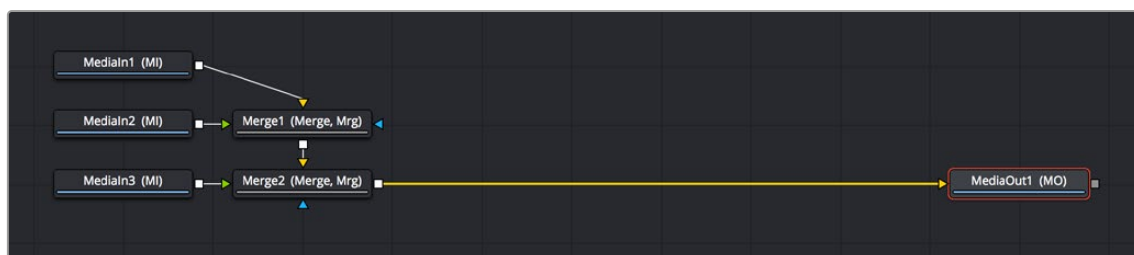
For more information on creating Fusion Clips, see *Chapter 65, "Getting Clips into Fusion."* in the DaVinci Resolve Reference Manual or *Chapter 3* in the Fusion Reference Manual.



A stack of clips to use in a composite (previous page), and turning that stack into a Fusion clip in DaVinci Resolve's Edit page (above)

In Fusion, each video clip is represented by a MediaIn in the Fusion page or a Loader in Fusion Studio.

In our example below, the MediaIn2 is video track 2, and MediaIn1 is video track 1. These two elements are composited using a Merge node (foreground over background, respectively). The composite of those two elements becomes the background to a second Merge. There is no loss of quality or precomposing when you chain Merges together. MediaIn3 represents video track 3 and is the final foreground in the node tree since it is the topmost layer.



The initial node tree of the three clips we turned into a Fusion clip

With this node tree assembled to mimic the video layers, we can focus the rest of this example on adding the nodes we'll need to each branch of this tree to create the green-screen composite.

Pulling a Green-Screen Key Using the Delta Keyer

To make this composite, you need to create transparency behind the newscaster. When working in a node tree, you must become accustomed to rearranging existing nodes to make room for new ones. You'll often move nodes off to the side or up above to make room for the additional nodes.



Creating space after the MediaIn nodes and selecting the second one in preparation for adding a node

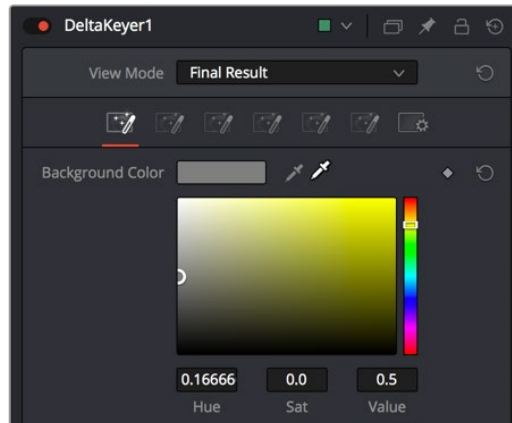
The DeltaKeyer node is the main tool used for green-screen keying. It attaches to the output of the node that represents the green screen—in our example, that is the MediaIn2 node. With the MediaIn2 selected, pressing Shift-Space opens the Select Tool dialog where you can search for and insert any node. Below we have added the DeltaKeyer after the MediaIn2 node but prior to being merged with the background.



Adding a DeltaKeyer node inline after the MediaIn2 node

The DeltaKeyer node is a sophisticated keyer that is capable of impressive results by combining different kinds of mattes and a clean-plate layer, but it can also be used very simply if the background that needs to be keyed is well lit. And once the DeltaKeyer creates a key, it embeds the resulting alpha channel in its output, so in this simple case, it's the only node we need to add. It's also worth noting that, although we're using the DeltaKeyer to key a green screen, it's not limited to keying green or blue only; the DeltaKeyer can create impressive keys on any color in your image.

With the DeltaKeyer selected, we'll use the Inspector controls to pull our key by quickly sampling the shade of green from the background of the image. To sample the green-screen color, drag the Eyedropper from the Inspector over the screen color in the viewer.



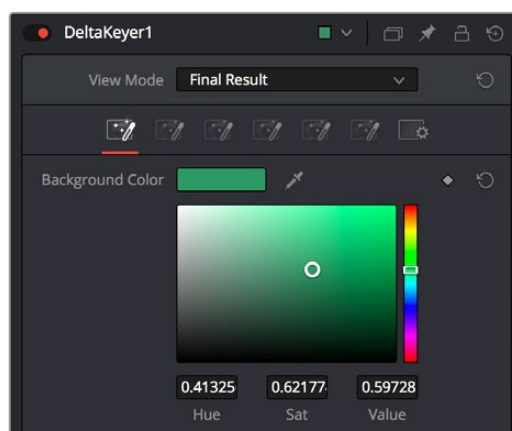
Dragging the Eyedropper to the viewer samples the screen color

As you drag in the viewer, an analysis of the color picked up by the location of the Eyedropper appears within a floating tooltip, giving some guidance as to which color you're really picking. Meanwhile, if viewing the Merge in a second viewer, we get an immediate preview of the transparency and the image we've connected to the background.

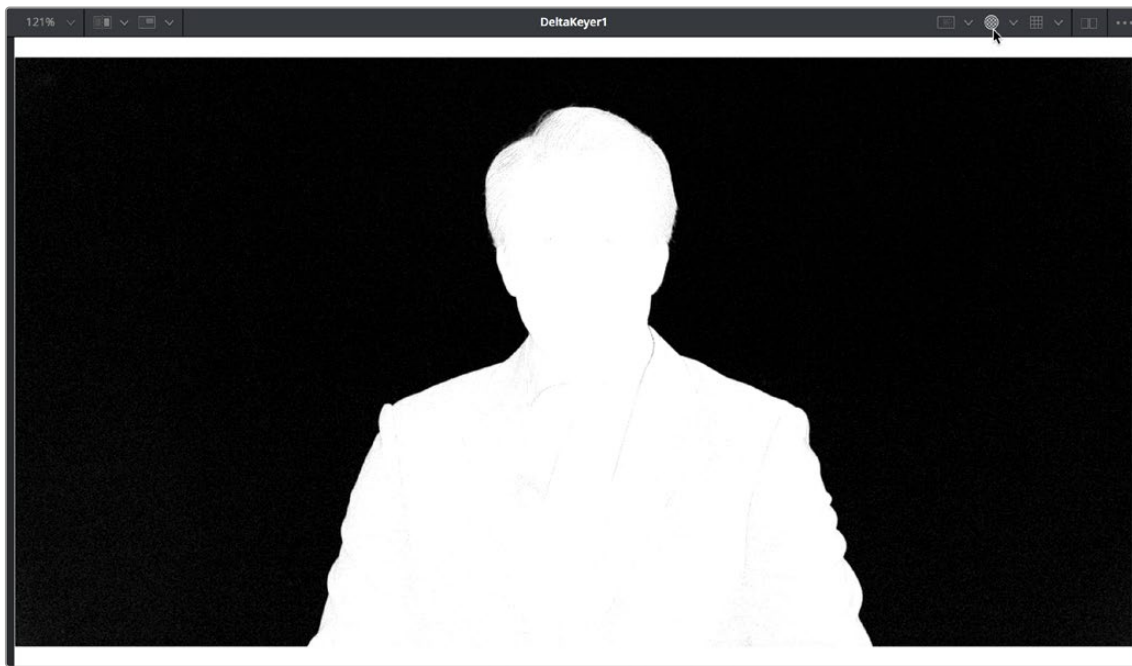


The original image (left), and after sampling the green screen using the Eyedropper from the Inspector (right)

When we're happy with the preview, releasing the pointer button samples the color, and the Inspector controls update to display the value we've chosen.



The DeltaKeyer Inspector updates with the sampled color



Loading the DeltaKeyer into the viewer and clicking the Color button to view the alpha channel being produced

No matter how good the composite may look, once you've selected the screen color to pull a key, you need to load the DeltaKeyer node into the viewer itself. This allows you to evaluate the quality or density of the alpha channel created by the key. Above the viewer, click the Color button in the viewer toolbar, or click in the viewer and press C to switch the viewer between the RGB color channels of the image and the alpha channel.

Black in a matte represents the transparent areas, while white represents the opaque areas. Gray areas represent semi-transparency. Unless you are dealing with glass, smoke, or fog, most mattes should be pure white and pure black with no gray areas. If a close examination of the alpha channel reveals some fringing in the white foreground of the mask, the DeltaKeyer has integrated controls for post-processing of the key and refining the matte. Following is a quick checklist of the primary adjustments to make.

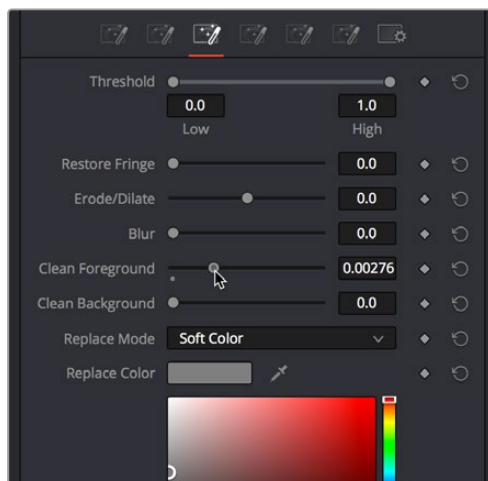
After making the screen selection with the Eyedropper, try the following adjustments to improve the key.

- Adjust the Gain slider to boost the screen color, making it more transparent. This can adversely affect the foreground transparency, so adjust with care.
- Adjust the Balance slider to tint the foreground between the two non-screen colors. For a green screen, this pushes the foreground more toward red or blue, shifting the transparency in the foreground.

Clicking the third of the seven tabs of controls in the DeltaKeyer Inspector opens up a variety of controls for manipulating the matte.

Initial adjustments in the matte tab may include the following parameters:

- Adjust the lower and upper thresholds to increase the density in black and white areas.
- Very subtly adjust the Clean Foreground and Clean Background sliders to fill small holes in the black and white matte. The more you increase these parameters, the more harsh the edges of your matte become.



Adjusting the Clean Foreground slider in the Matte panel of the DeltaKeyer controls

In this case, raising the Clean Foreground slider a bit eliminates the inner fringing we don't want, without noticeably compromising the edges of the key.



The original key (left), and the key after using the Clean Foreground slider (right)

With this accomplished, we're happy with the key, so we load the Merge1 node back into the viewer, and press C to set the Color control of the viewer back to RGB. We can see the graphic in the background, but right now it's too small to cover the whole frame, so we need to make another adjustment.



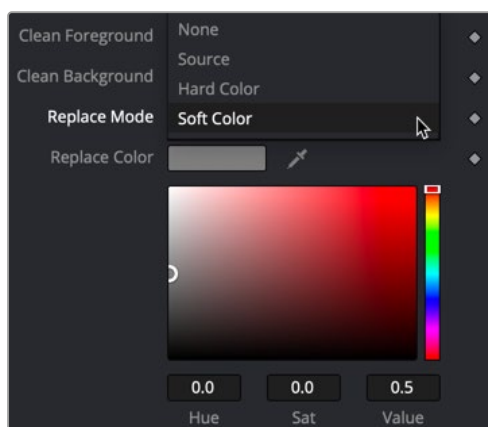
The final key is good, but now we need to work on the background

Dealing with Spill

The DeltaKeyer can handle any screen color that may bounce off the background and land on your subject. This bounce screen light is called spill. However, instead of using one node to handle the matte creation and the spill correction, consider disabling this in the DeltaKeyer and remove the spill using a separate color correction node.

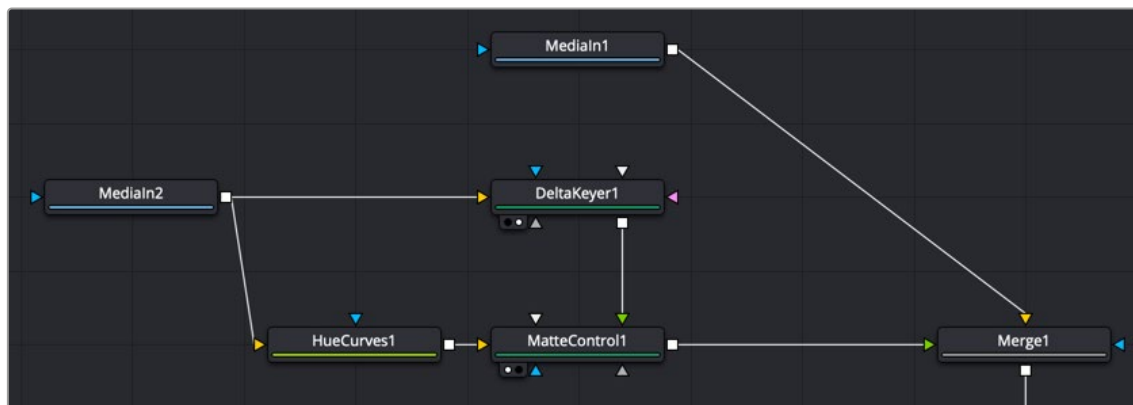
To disable spill suppression in the DeltaKeyer, do the following.

- Select the matte tab in the Inspector.
- From the Replace Mode drop-down menu, choose Source. This uses the color from the original source image, effectively disabling any spill correction in the DeltaKeyer.



The Replace Mode menu determines how spill suppression is handled in the DeltaKeyer

Spill can now be handled using a color correction node placed directly after the DeltaKeyer or branched from the original MediaIn or Loader node and combined with a MatteControl.



Branching the original image with one branch for the DeltaKeyer and a second for color correction

Masking a Graphic

Next, it's time to work on the top video track: the news graphic that will appear to the left of the newscaster. The graphic we will use is actually a sheet of different logos, so we need to cut one out using a mask and position it into place.



A graphic of multiple logos that must be cropped down to isolate just one

The easiest way to crop a MediaIn or Loader node is to add one of the mask shapes from the toolbar directly to it. Selecting the MediaIn or Loader node and clicking the Rectangle mask from the toolbar will crop, or mask off, the graphic.



Masking the logo using a Rectangle mask connected directly to a Merge node

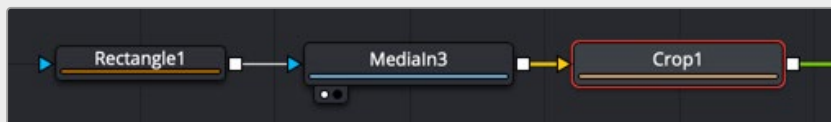
Now, all we need to do is to use the onscreen controls of the Rectangle mask to crop the area we want to use, dragging the position of the mask using the center handle, and resizing it by dragging the top/bottom and left/right handles of the outer border.

As an extra bonus, you can take care of rounded corners when masking a graphic by using the Corner Radius slider in the Inspector controls for the Rectangle mask to add the same kind of rounding.



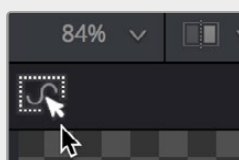
Moving and resizing the mask to fit our logo, and rounding the edges using the Corner Radius Inspector control

For a simple over-the-shoulder graphic, masking the image may be all you need to do, but masking an image does not change the actual dimensions of the graphic. It only changes the area you see. So, accurately positioning the graphic based on the center of the composite becomes more difficult, and any type of match moving would give incorrect results because the graphic has a different resolution than the background. To fix this resolution mismatch, you can place a Crop node after the MediaIn to change the actual dimensions of the graphic layer.



Adding a Crop node after the masked MediaIn to center the cropped logo on the background

With the Crop node selected, the viewer toolbar includes a Crop tool.



Selecting the crop tool in the viewer toolbar

You can crop the image by dragging a bounding box around it. Unlike a mask which creates a small window you view the image through, a crop effectively changes the resolution of the graphic to the crop bounding box size.



Dragging a bounding box using the Crop tool (left), and the cropped logo now centered on the frame (right)

NOTE: The Resize, Letterbox, and Scale nodes also change the resolution of an image.

At this point, we're all set to move the logo into place. Because the logo is the foreground input to a Merge, you can select the Merge2 node, load it into the viewer, and use the built-in Center X and Y controls or the onscreen controls to place the logo where you want it and make it a suitable size.



Placing the logo using the foreground input transform controls of the Merge2 node